Adaptive Running Plan Generator

Product Requirements Document — Version 1 (Couch-to-5K)

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1. Problem & Purpose

Existing couch-to-5K apps (e.g. the NHS app, Runna) are built around a fixed weekly template and, in Runna's case, a paid subscription. This project builds a personal, free alternative: a run-plan engine that generates a couch-to-5K schedule around the user's own available days, and adapts the plan when sessions are missed.

Primary purpose: a CV / portfolio project demonstrating algorithmic design (scheduling, constraint handling, plan adaptation) rather than CRUD or UI work alone.

Note: plan length is goal-specific. C25K runs 9 weeks; other distance goals (10K, 15K, half marathon, marathon), planned for later versions, will each have their own length — none of them are fixed to 9 weeks.

2. Audience

- Primary: the developer, as a personal training tool and a CV project.
- Secondary: anyone starting a couch-to-5K programme who wants a schedule that fits their own week rather than a fixed calendar.

3. Scope — Version 1

In scope:

- One goal only: **Couch-to-5K (C25K)** — a fixed 9-week run/walk interval progression.
- A plan generator that places sessions onto the user's chosen available weekdays, starting from a chosen start date.
- A replan function that adjusts the remaining schedule when sessions are missed (see Section 6).
- Automated tests (pytest) covering generation and replanning, including edge cases.
- Calendar export: a generated plan can be exported as an **.ics** file, importable into Google Calendar, Apple Calendar, or Outlook.

Out of scope for v1 (deferred):

- 10K, 15K, half marathon and marathon plans, and the recovery/speed/long session structure they require.
- Pace targets or pace-based speed work (effort-based cues only, e.g. "3 min hard").
- A front end / UI of any kind — v1 is a script or CLI producing an .ics file.
- Auto-detection of fitness level; the user simply selects C25K.
- Notes/journaling per session, and any calorie or step tracking.

3b. Tech Stack

- **Language:** Python (engine, tests, CLI). TypeScript may be used later if a web front end is built, but is out of scope for v1.
- **Testing:** pytest.

- **Calendar export:** the **ics** Python library (rather than hand-writing the .ics format), to keep focus on the scheduling logic.
- **Version control:** new GitHub repository, with a Python virtual environment (venv) for dependencies — consistent with previous projects.
- **Interface (v1):** command-line only — no front end. A UI (Figma design, then a web app) is deferred until after the engine and tests are complete; designing screens now would mean guessing at a UI for logic that doesn't exist yet.

4. Inputs & Outputs

Inputs (PlanRequest):

- **Goal** — fixed to C25K in v1.
- **Start date** — the date the plan begins.
- **Available days** — the weekdays the user can run (must supply at least 3, since C25K runs 3x/week).
- **Max session minutes** — used to sanity-check that a session fits the user's available time (C25K sessions are ~20–35 minutes including warm-up/cool-down).

Output (Plan):

- An ordered list of **Sessions**, one per scheduled run, each with: a date, a week number (1–9), a type (walk/run interval structure), a list of segments (e.g. "run 1 min, walk 90 sec, x8"), and a status (planned / done / missed).
- An .ics file representing the current plan, regenerated after any replan.

5. Core Rules (v1 — C25K)

- The plan always follows the standard 9-week C25K run/walk progression (fixed week-by-week structure; not user-configurable in v1).
- Exactly 3 sessions are scheduled per week, placed on the user's chosen available days.
- Sessions are spaced out across the week where possible, rather than clustered on consecutive days, to allow recovery.
- The plan is a pure function of its inputs: given the same request, it always produces the same schedule (no hidden state or randomness).

6. Missed-Session Handling (Replan)

Behaviour is modelled on how Runna handles this: a single missed session is not crammed in later, but persistent misses trigger a bigger adjustment (Runna's own "Plan Realignment" feature appears once more than three sessions, or a full week, have been missed).

Rules for v1 — three tiers:

- **1 miss in a week:** mark it "missed", then shift the remaining "planned" sessions in that week onto the next available days from today. The week is not extended for a single miss.
- **2 misses in a week (week A):** shift the remainder the same way as a single miss, but flag the week as at risk. No week is repeated yet on its own.
- **Escalation — a miss in the week right after an at-risk week (week B):** look at week B's first session. If that first session was missed, the user never properly started week B, so **week A is repeated**, followed by

a fresh restart of week B — the plan grows by two weeks. If week B's first session was completed and a later one was missed instead, **week B is repeated** in place — the plan grows by one week.

- **3 or more misses in a single week (any week):** that week is repeated immediately, regardless of the weeks before or after it — the plan grows by one week.
- In every repeat case, any sessions in the repeated week that had not yet happened are dropped; the repeat replaces them rather than running alongside them.
- The week-by-week C25K structure itself never changes; only which calendar dates each week's sessions fall on, and how many weeks the plan contains.
- Replanning only ever affects sessions that are still "planned" — completed or already-missed sessions are never altered, only repeated going forward.
- Replanning is idempotent: running it again on an already-replanned schedule with no new misses makes no further changes.

7. Acceptance Criteria

Each of these is intended to become a pytest test case:

- Given a start date and 3 available weekdays, **generate_plan** returns exactly 27 sessions (9 weeks x 3), each falling on one of the chosen weekdays.
- Given only 2 available weekdays, **validate_request** rejects the request (C25K needs 3 sessions/week).
- Marking one session as "missed" and calling **replan** reschedules only the remaining sessions in that week, onto the user's available days, without changing sessions in later weeks.
- Marking all 3 sessions in a week as "missed" causes that week to repeat; all later weeks shift forward by one, and the plan's total length grows by one week.
- Marking 2 sessions missed in week A, then the first session of week B missed, causes week A to repeat followed by a fresh week B — the plan grows by two weeks.
- Marking 2 sessions missed in week A, then a later (non-first) session of week B missed, causes week B alone to repeat — the plan grows by one week.
- Calling **replan** a second time with no new misses since the first call produces an identical plan (idempotence).
- A completed ("done") session is never modified by **replan**.
- The exported .ics file contains one calendar event per planned session, each with the correct date and a description of the run/walk structure.

8. Build Order

- **1. Data model** — PlanRequest, Session, Segment (done — see plan_model.py).
- **2. validate_request** — reject invalid requests (e.g. too few available days).
- **3. generate_plan** — build the fixed 9-week C25K schedule onto available days.
- **4. Tests** for 2 and 3, covering the acceptance criteria above.
- **5. replan** — single-miss shifting, then repeated-miss week restart, with tests.
- **6. .ics export** — convert a Plan into a calendar file.
- **7. CLI entry point** — collect inputs, run the engine, write the .ics file.

9. Definition of Done — Version 1

- generate_plan and replan are implemented and covered by passing pytest tests for all acceptance criteria in Section 7.
- A plan can be exported to a valid .ics file that opens correctly in at least one calendar app.
- Code is committed to GitHub with a README explaining the project, the rules above, and a disclaimer that the schedule is a heuristic, not medical or coaching advice.

Note: progression numbers and scheduling rules above are simplified for a first build and are not medical or coaching advice.